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REVIEW PAPER

Artificial intelligence for prognostics and health management in critical machinery: a systematic review

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Abstract

This study presents a systematic literature review on the integration of Artificial Intelligence (AI) and Machine Learning (ML) techniques into Prognostics and Health Management (PHM), which integrates condition monitoring, diagnosis, and prognostics to support predictive maintenance decisions. Although several review studies have addressed artificial intelligence-based predictive maintenance, a systematic and structured analysis capable of explicitly identifying research gaps and unexplored combinations within PHM remains limited. Motivated by the increasing relevance of predictive maintenance in Industry 4.0, the review examines how modern AI architectures have been applied to Health State (HS) estimation and Remaining Useful Life (RUL) prediction, which represent, respectively, the current condition of the system and the expected time until functional failure, across critical machinery. To address this gap, the study employs an integrated review framework combining the Systematic Search Flow (SSF), the Morphological Matrix (MM), and the Cross-Consistency Matrix (CCM), which together structure the literature selection, classification of research dimensions, and identification of unexplored variant interactions, enabling a rigorous and multidimensional analysis beyond conventional descriptive reviews. From an initial dataset of 692 publications, 18 primary studies were selected and classified according to four key dimensions: PHM aspect, AI/ML technique, analyzed element, and data-processing interface. The results reveal a strong concentration of research on CNN- and LSTM-based RUL prediction, where Convolutional Neural Networks (CNN) and Long Short-Term Memory networks (LSTM) are used to model spatial and temporal degradation patterns, for bearings, while complex systems, multimodal sensing, embedded intelligence, and IoT-, IIoT-, and AIoT-based interfaces, referring respectively to the Internet of Things, the Industrial Internet of Things, and the integration of artificial intelligence capabilities at the device or edge level, remain largely unexplored. The CCM reveals that only 48% of the 96 possible variant interactions are represented in current literature. This quantitative finding highlights substantial unexplored research opportunities, particularly with respect to real-world deployments, distributed processing architectures, and the development of methodological frameworks for model verification and field validation. Overall, the findings provide a consolidated view of the state of the art and outline a clear research agenda for advancing AI-enabled PHM in industrial environments. By quantitatively mapping these gaps and synthesizing existing approaches, this review contributes a structured perspective that supports the development of practical, scalable, and industrially deployable PHM solutions. As machinery becomes increasingly connected and autonomous, bridging the gap between high-performance predictive models and operational deployment will be essential for realizing the full potential of intelligent maintenance strategies.

Keywords Prognostics and health management · Machine learning · Industry 4.0 · Systematic literature review

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1 Introduction

The industrial revolutions were shaped by three main drivers: knowledge, experimentation, and innovation. They cannot be reduced merely to the materials, methods, machines, and energy sources employed [48]. The first industrial revolution was marked by the shift to steam power, symbolized by the engine invented by James Watt in 1765 [67]. Approximately one hundred years later, the second industrial revolution emerged, characterized by mass production lines and electrification. In 1969, the third industrial revolution began with the development of the first programmable logic controller, marking the advent of industrial automation [31].

In 2012, the concept of Industry 4.0 was introduced in Germany and was soon adopted by other developed countries, including the United Kingdom, Japan, France, and the United States of America [61]. According to Coelho [15] and Wolfert et al. [79], the smart industry (Industry 4.0) relies on a wide set of emerging technologies, such as the Internet of Things (IoT), Big Data, cloud computing, and artificial intelligence. The implementation of digitally enabled machines and automated production systems has led to a substantial increase in productivity levels [58].

In parallel with the industrial revolutions, maintenance practices have evolved to address increasingly demanding operational requirements, giving rise to new approaches, techniques, and theories. The correlation between maintenance strategies and the industrial revolutions was examined in detail by Poór et al. [51], who associated the four industrial revolutions with reactive, planned, productive, and predictive maintenance types, respectively. This evolution is closely linked to improvements in Overall Equipment Effectiveness (OEE) across each period.

Corrective maintenance is generally based on visual inspections and is applied reactively, typically resulting in OEE values below 50%. In contrast, predictive maintenance leverages sensor data and advanced analytics, making it possible to achieve OEE values above 90% [51]. Predictive maintenance is inherently knowledge-based, whether at the component, system, or organizational level. According to Nadj et al. [45] and Lüttenberg et al. [39], it enables better planning of maintenance activities, reduces equipment downtime, improves operational efficiency, and can significantly reduce costs.

Among predictive maintenance strategies, Prognostics and Health Management (PHM) stands out as one of the most comprehensive and widely recognized approaches. PHM integrates health-state monitoring and Remaining Useful Life (RUL) prediction into a unified framework [49]. Its effective implementation depends on the availability of large volumes of reliable data and information. Recent studies by Allal et al. [4], Cabrera et al. [11], and Xu et al. [81] have explored the integration of Machine Learning (ML) and Artificial Intelligence (AI) techniques within PHM architectures. However, these contributions are mostly based on standardized datasets and do not include real-time data acquisition.

Currently, most high-cost equipment is extensively instrumented, for instance, CNC machines are equipped with humidity, temperature, and vibration sensors, commercial vehicles such as trucks and buses are equipped with CAN-bus systems, which transmit data such as engine speed, ambient temperature, and air pressure, among many others, and agricultural machinery, such as sugarcane harvesters, transmit data through internal bus systems containing signals from various points, including hydraulic pressure, engine speed, cutter implement rotation, and vehicle speed [3, 28, 38].

However, certain limitations in operational environments hinder the full utilization of this data. These include high data transfer latency, limited network coverage in remote areas, application-specific constraints, and the availability of installed sensors. These factors reveal the under-utilization of valuable information for analyses beyond those conducted by the equipment's control system. Thus, by understanding the state of the art in the integration of these domains, it is possible to identify trends, observe technological evolution, and recognize opportunities for novel combinations that have yet to be explored.

Recent review studies have investigated the application of artificial intelligence and data-driven methods to predictive maintenance and Prognostics and Health Management. Existing surveys commonly focus on specific algorithms, application domains, or component types, such as bearings, batteries, or rotating machinery, and often provide qualitative taxonomies or performance-oriented comparisons. While these reviews offer valuable insights into algorithmic trends and benchmark results, they generally lack a structured framework capable of systematically classifying research dimensions and quantitatively identifying unexplored combinations across PHM aspects, system types, and processing interfaces. As a result, the identification of research gaps is frequently based on narrative assessment rather than explicit analytical mapping.

In contrast to existing review studies, which primarily organize the literature through narrative discussions, algorithm-centered taxonomies, or performance-based comparisons, the present review adopts a structurally driven analytical approach. By integrating the Systematic Search Flow (SSF) with the Morphological Matrix (MM) and the Cross-Consistency Matrix (CCM), this study enables a formal decomposition of the PHM problem space into dimensions and variants, followed by a quantitative mapping of their interactions. This approach allows not only

the identification of dominant research trends, but also the explicit quantification of unexplored combinations that remain absent from current literature, thereby providing a level of analytical rigor and transparency that is generally not achieved in conventional review methodologies.

1.1 Contributions of this study

This study provides several relevant contributions to the intersection between Prognostics and Health Management (PHM) and advanced Artificial Intelligence (AI) techniques. The main contributions are summarized as follows:

i. *A focused and structured systematic literature review* on the integration of PHM with contemporary AI and Machine Learning (ML) methods, highlighting how modern architectures have been applied to Health State (HS) estimation and Remaining Useful Life (RUL) prediction.
ii. *The application of a multi-stage methodological framework* combining the Systematic Search Flow (SSF), Morphological Matrix (MM), and Cross-Consistency Matrix (CCM). This integrated approach provides a comprehensive, transparent, and reproducible structure for analyzing complex variant interactions within the domain.
iii. *A morphological classification of the most relevant studies* based on four key dimensions: PHM aspect, AI/ML technique, analyzed element, and data-processing interface. This classification enables a deeper understanding of dominant research patterns and methodological tendencies.
iv. *Identification and quantification of unexplored research combinations*, revealing that only 48% of the theoretically possible variant interactions are covered in current literature. Notably, the analysis exposes the limited presence of IoT, IIoT, and AIoT in PHM-oriented applications.
v. *A synthesis of methodological artifacts found in state-of-the-art studies*, including subsystem identification strategies, data preprocessing approaches, feature extraction methods, and evaluation metrics. This synthesis offers a practical reference for future PHM implementations.
vi. *A consolidated research agenda* pointing to promising directions such as real-world multi-sensor deployments, embedded intelligence using AIoT, PHM in complex machinery with limited connectivity, and the development of verification and validation methodologies for PHM systems.

Together, these contributions provide a structured and evidence-based understanding of how AI techniques have been incorporated into PHM and highlight opportunities for advancing predictive maintenance in industrial and critical operational systems.

2 Theoretical background

The following sections present a comprehensive overview of key topics that underpin this study. Including maintenance management fundamentals and the main types of maintenance strategies: Corrective Maintenance (CM), Preventive Maintenance (PM), and Predictive Maintenance (PDM), with special emphasis on Prognostics and Health Management (PHM). Additionally, specific challenges related to maintenance in critical applications are discussed. Finally, the integration of Artificial Intelligence (AI) and Machine Learning (ML) techniques is explored, including the application of advanced neural network architectures such as Deep Neural Networks (DNNs).

2.1 Maintenance management

By definition, maintenance is the collection of activities required to retain or restore a given item to its specified desired condition (US-DoD, 1998). The way these activities are conducted or planned is directly related to how failures are addressed at their onset, during their progression, and at their conclusion. This relationship is commonly represented in the literature by the *P-F* curve, where *P* denotes the potential failure, the point at which degradation begins, and *F* represents the functional failure [76]. According to Josebeck and Gowtham [33], the *P-F* curve is a theoretical model that describes the operational state of a system from full functional capacity to complete failure over a usage period, capturing all intermediate stages of degradation. Some authors, such as Nowlan and Heap [47] and Moubray [44], define point *F* as the condition in which the equipment can no longer meet specified performance requirements, whereas point *P* is identified as the moment when system degradation begins, even if not immediately critical [33].

Maintenance strategies have progressively evolved from reactive approaches toward data-driven methodologies aimed at improving reliability, availability, and cost efficiency of industrial assets. Traditional preventive maintenance policies, based on fixed schedules or usage intervals, often lead to either unnecessary interventions or unexpected failures, motivating the development of predictive maintenance strategies.

Predictive maintenance seeks to anticipate failures by monitoring the condition of systems and components during operation, enabling maintenance actions to be planned based on actual degradation rather than predefined intervals.

However, conventional condition-based approaches are frequently limited to threshold-based indicators or isolated signal analysis, which restrict their ability to capture complex degradation mechanisms and interactions among subsystems.

In this evolutionary framework, PHM emerges as an integrated framework that extends predictive maintenance by combining condition monitoring, diagnostics, and prognostics to support informed maintenance decision-making. The increasing availability of sensor data and computational resources has further enabled the adoption of artificial intelligence and machine learning techniques to enhance PHM capabilities, motivating the growing interest in AI-driven approaches analyzed in this review.

This evolution from schedule-based maintenance toward data-driven PHM frameworks provides the conceptual foundation for the systematic review and analytical methodology presented in the subsequent sections. Based on these concepts, Fig. 1 schematically illustrates the hierarchical relationship between maintenance strategies, along with their specific approaches and associated methods.

2.1.1 Corrective maintenance (CM)

Corrective Maintenance (CM) encompasses all actions required, as a result of a failure, to restore a system or component to its required functional state [75]. It is also referred to as reactive maintenance [69]. According to Chalifoux and Baird [12], this approach demands less planning effort but is associated with higher costs due to unplanned downtime, increased expenses for repairing peripheral systems, and the potential for collateral damage. CM can be classified into two main categories:

- Planned corrective maintenance: results from a deliberate decision to allow a system or component to operate until failure, commonly referred to as "run-to-failure." This strategy is often applied to low-cost components where failure has minimal operational impact, such as light bulbs [22].
- Unplanned corrective maintenance: occurs in response to unexpected failures that cause a reduction in system performance or a complete interruption of function. These failures typically arise due to randomness or insufficient monitoring.

2.1.2 Preventive maintenance (PM)

According to Mobley [42], Preventive Maintenance (PM) is a time-dependent and periodic approach, based either on continuous counting or accumulated usage hours. Therefore, its most relevant metrics relate to failure indices and time-based measures such as Mean Time Between Failures (MTBF) and Mean Time To Failure (MTTF). PM includes routine practices such as inspections, testing, and repairs aimed at detecting, reducing, or eliminating failures [16]. While this type of maintenance tends to reduce failure-related costs, it may increase component costs due to fixed replacement intervals, disregarding the individual randomness of item degradation [12]. PM can be further divided into the following main categories:

- Time-Based Maintenance (TBM): A straightforward technique that relies solely on the service time of components and historical failure data. Although easy to implement, TBM often results in premature replacement [32].

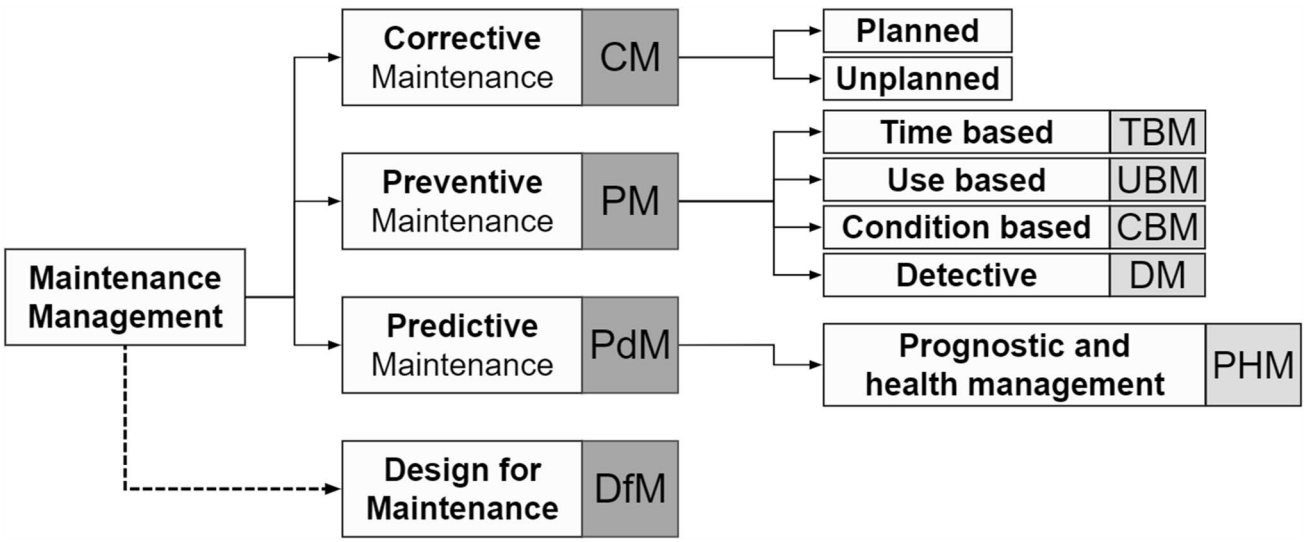


Fig. 1 Maintenance management strategies

- Usage-Based Maintenance (UBM): Similar to TBM, but intervals are based on usage-related factors beyond time alone, such as distance traveled or hours of operation [23].
- Detective Maintenance (DM): Although some authors, such as de Almeida Júnior and Parra [20] and Bezerra [7], classify DM as a standalone strategy, it is better characterized as an initial stage of Predictive Maintenance (PDM). DM focuses on identifying subtle or hidden faults and assessing whether the system is fully operational, thus supporting early-stage fault diagnosis that might not be immediately perceptible to an operator.
- Condition-Based Maintenance (CBM): While [20] suggest that PDM and CBM are largely similar, some distinctions remain. CBM does not involve predictive analysis but instead triggers maintenance actions only when real-time data indicate an anomaly. A more precise definition was provided by Horner et al. [26], Raheja et al. [54], and Mohsen [43], that describe CBM as a strategy initiated solely upon the detection of specific conditions through real-time data analysis. In this study, CBM is considered a subset of PM.

2.1.3 Predictive maintenance (PDM)

Predictive Maintenance (PDM) relies on the continuous monitoring and assessment of a system's current condition through the use of specialized resources, such as sensors and system-to-system communication [42]. A more comprehensive definition is presented by Ippolito et al. [29] and Crespo Marquez et al. [18], who describe PDM as a means of identifying existing patterns through large-scale data analysis in order to predict or prevent potentially damaging conditions. According to Sullivan et al. [69], as of the year 2000, maintenance practices in the United States were composed of over 55% corrective, 31% preventive, and only 12% predictive maintenance.

Furthermore, Sullivan et al. [69] report that shifting from corrective to preventive maintenance can result in estimated cost reductions between 12% and 18%. Transitioning from preventive to predictive maintenance may yield an additional 8% to 12% reduction. However, such gains often require substantial investments in diagnostic equipment and personnel training.

Among the available PDM approaches, Prognostics and Health Management (PHM) stands out as the most comprehensive and widely recognized for its benefits. According to Biggio and Kastanis [8], PHM utilizes information extracted from systems to assess the current health state of components and guide maintenance operations as needed. Additionally, Pecht [49] states that PHM originated from the concept of determining the health status, or degree of degradation, of an asset with a specified level of confidence, enabling the prediction of required interventions.

2.2 Prognostics and health management (PHM)

PHM aims to prevent unforeseen failures that may compromise system performance and to protect system integrity [1]. According to Compare et al. [17], this technology enables, in principle, condition monitoring, estimation of Remaining Useful Life (RUL), and the prediction of catastrophic failures. PHM is founded on the integration of computational methods and signals from structures, components, and systems, facilitating anomaly detection, diagnosis execution, fault evolution prediction, and RUL estimation. Furthermore, Boulzazen et al. [10] and Teixeira et al. [72] argue that PHM goes beyond CBM by utilizing collected data not only to assess the current condition of the monitored equipment but also to extrapolate this information over time. Through physics-based models, PHM estimates the degradation progression and predicts the future system state with a certain level of confidence. Biggio and Kastanis [8] define PHM as a convergence of CBM and PM, encompassing both fault identification and RUL estimation.

In 2017, the IEEE Standard 1856-2017 was published as a recommended framework for implementing PHM in electronic systems. The standard defines prognostics as the process of predicting RUL based on fault progression and degradation levels, estimating how much longer a given system can perform its specified function. Health management, in this context, is considered the decision-making process that uses the analyzed data and RUL predictions to support maintenance actions.

In summary, PHM is a relatively recent and evolving methodology, currently considered the most established approach for implementing PDM in industrial environment. Although IEEE 1856-2017 focuses on electronic systems, it explicitly states that the recommendations can be extended to mechanical, electromechanical, or structural systems. According to Kwon et al. [36], PHM consists of four core dimensions: sensing, diagnosis, prognosis, and health management.

Sensors enable the detection of anomalies, changes in operating conditions, and tracking of machine performance or degradation states. Based on the collected data, anomalies associated with potential failure modes can be identified, supporting the diagnosis stage. By analyzing, modeling, and projecting these data, alongside domain knowledge about possible failures, prognostic information can be derived within a certain confidence interval. These three stages together inform effective asset management decisions. This need becomes even more evident when dealing with critical systems.

The concept of what constitutes a critical piece of equipment may vary depending on context. According to Akinyemi [2], critical machines are those that perform essential functions within an industrial environment, where their unavailability directly impacts the production timeline of a given item. In a similar vein, Iaksch [28] specifically identify harvesting machines as critical assets in the agricultural sector. For Periyasam and Dhanasekaran [50], critical applications include hospital equipment such as mechanical ventilators, infusion pumps, and other life-support systems.

A critical system or piece of equipment can be understood as one that operates in remote or hard-to-reach locations, presents limited communication capabilities, causes process interruption, entails financial losses, or poses a risk to human life and operator or user safety when unavailable or malfunctioning. These characteristics underscore the need for proactive monitoring and maintenance strategies, particularly in contexts where failures are difficult to detect or respond to in a timely manner. Within this context, maintenance emerges not only as a technical practice, but also as an area closely linked to cognitive science domains, especially in terms of decision, making under uncertainty, diagnostic reasoning, and human–machine interaction. Consequently, the identification and management of critical equipment is a key element in the implementation of effective Prognostics and Health Management (PHM) solutions.

In this study, PHM is considered as an integrated framework that combines sensing, diagnosis, and prognosis to support maintenance decision-making, serving as the conceptual foundation for the subsequent literature classification and analysis.

2.3 Artificial intelligence and machine learning

The field of Artificial Intelligence (AI) goes beyond merely identifying patterns; it seeks to create intelligent entities. Its development began shortly after World War II, in 1956, and it now encompasses a wide range of knowledge domains [68]. According to Khanzode and Sarode [35], AI focuses on methods for solving symbolic and non-algorithmic problems, and it can be understood based on its various application areas. AI encompasses a broad range of approaches and application domains, including cognitive systems, learning-based methods, and intelligent agents applied to perception, decision-making, and control tasks in industrial environments.

Cognitive applications of Artificial Intelligence play an essential role in various domains, and in the context of maintenance, they are particularly relevant for supporting complex decision-making processes, diagnostics, and prognostics under uncertainty. Among these cognitive applications, a particular emphasis is placed on the subfield of Machine Learning (ML), which is capable of adapting to, detecting, and extrapolating patterns. The term Machine Learning was coined by Samuel [59] in reference to his work on the first self-learning program for the game of checkers, where he defined ML as the field of study that enables computers to learn without being explicitly programmed. According to Das et al. [19], four distinct types of ML are widely recognized today: Supervised Learning, Unsupervised Learning, Semi-Supervised Learning, and Reinforcement Learning.

Additionally, recent publications have increasingly discussed Deep Learning, which Jakhar and Kaur [30] define as a subset of ML that uses computational models and algorithms inspired by the biological architecture of neural networks. In essence, when the brain receives new information, it attempts to compare it to previously known patterns. The term *deep* refers to the number of layers in the neural network, which enables the system to handle large amounts of data, particularly unstructured data, with high effectiveness.

The main ML methods can be summarized as follows:

- Supervised Learning: A learning process based on comparing a computed output with an expected output. The measured error is used to adjust the algorithm and reduce prediction discrepancies [19]. According to Nasteski [46], this method requires a large amount of labeled data, typically split into training and testing sets, and is widely used for classification and regression tasks.
- Unsupervised Learning: Commonly used to uncover inherent data structures without labeled outputs. It identifies similarities and clusters information into categories [73].
- Semi-Supervised Learning: Positioned between the supervised and unsupervised paradigms, this approach does not require as much labeled data as supervised learning, nor does it rely entirely on the pattern discovery required by unsupervised methods [55].
- Reinforcement Learning: This type of learning seeks to determine the optimal sequence of actions to maximize expected numerical reward. As described by Sutton and Barto [70], the agent is not explicitly instructed on what actions to take. According to Das et al. [19], positive outcomes result in rewards, while negative outcomes trigger penalties, enabling the agent to learn the best strategy through exploration.
- Deep Learning: In simplified terms, this refers to the combination of multiple sequential neural network layers-referred to as Deep Neural Networks (DNNs). Input data passes through multiple transformations across these layers until the final output is reached, which may be a classification or extracted feature [14]. DNN techniques are currently among the most advanced and

rapidly evolving in the field. Their internal structure and key functions will be addressed in detail in the following section.

2.3.1 Advanced neural network techniques – DNN

Research and development in Artificial Intelligence (AI) have gained significant relevance and scope due to advances in data processing capabilities, cloud services, and data availability. According to Shrestha and Mahmood [63] and Chen and Ran [14], Deep Neural Networks (DNNs) consist of multiple layers of interconnected nodes and can be applied to solve problems across various domains. The most prominent techniques within this field, each with practical applications, include Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Generative Adversarial Networks (GAN), as illustrated in Fig. 2.

- Convolutional Neural Networks (CNN): According to Gu et al. [25], there are currently various CNN architectures, generally composed of five basic layers: an input layer, a convolutional layer, a pooling layer, a fully connected layer, and an output layer. CNNs are capable of learning complex hierarchical features at different levels of abstraction, making them highly effective for pattern recognition, classification, and segmentation tasks. Despite their advantages in automatic feature extraction, CNNs are characterized by complex architectures, which can be difficult to maintain and computationally intensive [74].
- Deep Belief Networks (DBN): As noted by Li et al. [37], a standard DBN is composed of multiple Restricted Boltzmann Networks (RBNs), in which input data feed a visible layer that communicates with a hidden layer through weighted connections. DBNs are known for their ability to automatically extract high-level features and learn effective, hierarchical data representations, proving especially useful for pattern recognition, classification, and data generation tasks.
- Recurrent Neural Networks (RNN): Designed to handle sequential data where temporal order and dependencies matter, RNNs differ from traditional neural networks by including feedback loops that retain information through hidden states. These connections influence future time steps. However, they may suffer from vanishing or exploding gradient problems, making it difficult to learn long-term dependencies, an issue addressed by variants such as LSTM and GRU. According to Tyagi and Abraham [73], RNNs are advantageous due to their reduced number of parameters, which lowers computational cost, but they can be challenging to train.
- Long Short-Term Memory (LSTM): A specialized RNN variant designed to address problems where long-term temporal dependencies are critical. LSTMs excel in prediction, language translation, and sequence generation tasks. [62] describes them as an evolution of RNNs incorporating non-linearities. Tyagi and Abraham [73] explain that LSTMs contain memory cells capable of storing information over long periods, although they may be prone to overfitting.
- Generative Adversarial Networks (GAN): As explained by Booyse et al. [9], GANs consist of two adversarial models: a generator and a discriminator. The generator produces synthetic data samples, while the discriminator evaluates their authenticity by distinguishing them from real samples. Training GANs is notably difficult, as the generator attempts to deceive the discriminator, which in turn tries to improve its detection accuracy. This adversarial process leads to a continuous refinement,

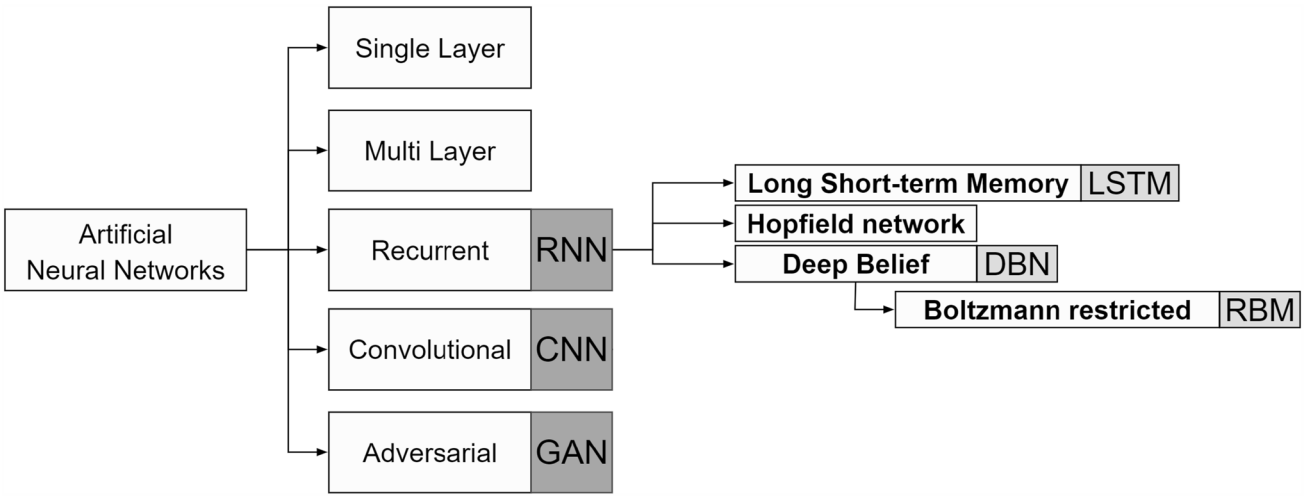


Fig. 2 Neural network structures. Source: Adapted from Tyagi and Abraham (2022)

ultimately resulting in synthetic samples that resemble real data.

3 Data in industry 4.0

In the context of Industry 4.0, characterized by the integration of *Internet of Things* (IoT) and *Big Data*, a large volume of data is continuously generated and collected from various sources, such as production, transportation, and after-market activities. These data hold significant value for corporate intelligence. However, the constant generation of massive datasets presents major challenges in terms of processing [56].

According to Wang et al. [78], consolidated data must encompass both large-scale historical datasets and data collected from the smallest connected components, with equal efficiency and speed. In response to these challenges, various structural alternatives have been developed, including *Cloud Computing*, *Edge Computing*, *Fog Computing*, among others. From a broader perspective, these architectures can be organized as illustrated in Table 1, comprising the equipment level, followed by the edge layer, and ultimately the cloud layer.

The choice of architecture in a given project is guided by its boundary conditions. According to Sisinni et al. [64], the most common evaluation metrics include real-time performance, energy efficiency, security, and privacy. Premsankar et al. [52] emphasize that data latency is the critical factor. Recent studies by Tank and Gandhi [71], Barakat et al. [5], and Wang et al. [78] have compared these different architectures, and a synthesis of their findings is presented in Table 1.

One of the pillars of Industry 4.0 development is the integration of cyber-physical systems, and the most visible element of this transformation is the proliferation of IoT devices. According to Tank and Gandhi [71], IoT devices generate massive amounts of data from sensors installed in various locations, enabling the interconnection of systems. The integration of machines, devices, and intelligent systems facilitates communication and information exchange within companies [60].

Table 1 Comparison among data processing architectures and their implications for PHM data latency, scalability, and deployment

	Cloud computing	Edge computing	Fog computing
Architecture	Centralized	Decentralized	Decentralized
Data processing location	External server	Near the data source	Local network
Latency	↑	↓	↓
Processing capacity	↑	Medium	↓
Scalability	Easy	Medium	Difficult
Average response time	Minutes	Milliseconds	Seconds
Bandwidth usage	↑	↓	Medium
Security	↓	↑	Medium

Source: Adapted from Tank and Gandhi [71], Barakat et al. [5], Wang et al. [78]

According to Iaksch [28], machinery can be equipped with sensors and communication devices to enable a smart environment. The benefits of smart systems are becoming a reality in industry, with various sectors increasingly adopting the technology.

A projection by Ketu and Mishra [34] estimates that more than 70 billion IoT devices will be deployed globally by 2025, highlighting the vast potential for system adoption in the coming years. Additionally, the integration of cloud computing and edge processing allows real-time data processing, enabling timely and decentralized decision-making. As IoT ecosystems evolve, concerns related to cybersecurity, data interoperability, and system scalability are also becoming critical areas of research and development.

3.1 IoT, IIoT and AIoT interfaces and applications

The Internet of Things is often portrayed as a disruptive solution to everyday problems such as public transportation and pollution monitoring. However, specific applications impose distinct requirements. In this context, Sisinni et al. [64] highlights the subdivision known as the Industrial Internet of Things (IIoT), which is applied in industrial communication domains to promote greater process efficiency and more sustainable production.

- Internet of Things (IoT): The term "IoT" was first introduced in 1997 and is defined as "a network of items connected to the Internet that generate and transmit data from sensors." [41].
- Industrial Internet of Things (IIoT): IIoT is a subdivision of IoT tailored to meet industrial requirements. It enables the monitoring, control, and intelligent analysis of data in industrial production environments, leading to increased efficiency and improved product quality [56].
- Artificial Intelligence of Things (AIoT): AIoT refers to the integration of Artificial Intelligence (AI) capabilities into Internet of Things (IoT) systems. While IoT enables connectivity and data acquisition, AI empowers devices to perform intelligent tasks such as learning, reasoning, and real-time decision-making. AIoT is fundamental to enabling autonomous and adaptive systems in applications such as predictive maintenance, smart manufacturing, and intelligent transportation [82].

In summary, IoT, IIoT, and AIoT systems are all capable of collecting and transmitting data. However, the data generated by these technologies is generally transmitted to high-performance computing platforms, where it undergoes advanced analysis to support complex decision-making processes. In the case of AIoT, this integration extends further by embedding artificial intelligence at the edge or within the system itself, enabling real-time processing and autonomous responses without relying solely on centralized infrastructure.

4 Systematic literature review

The development of the systematic literature review was based on well-established methods for identifying, evaluating, and interpreting published and peer-reviewed literature within the field of interest. This process followed the principles of systematic review methodology, morphological analysis, and the cross-consistency assessment matrix, as developed by Ferenhof and Fernandes [21], Ritchey [57], and Zwicky [84], respectively. At the conclusion of this process, the dispersed knowledge on the topic is consolidated to provide insight into research gaps and current trends in the literature.

Table 2 Papers collection for systematic literature review

Database	Search string	Restrictions	Results
ACM Dig. library	[[All: "prognostics health management"] OR [All: phm]] AND [[All: "artificial intelligence"] OR [All: ai]]	Research article AND journal AND english	212
Scopus	TITLE-ABS-KEY((phm) OR (prognostics AND health AND management) AND (artificial AND intelligence)OR(ai))	Research AND english	114
IEEE Xplore	("All Metadata":"prognostics health management" OR "All Metadata":phm) AND ("All Metadata":"artificial intelligence")	Journal AND english	65
Engineering village (COMPENDEX)	((((((("phm") OR ("prognostics health management")) WN ALL) AND (((("artificial intelligence") OR ("ai")) WN ALL))) AND (((cpx) WN DB) AND (ja WN DT)))	Journal AND english	144
ScienceDirect	(("phm") OR ("prognostics health management")) AND (("artificial intelligence") OR ("ai"))	Research articles AND english AND engineering	157
Total	–	–	692

4.1 Systematic review procedure

As mentioned, the material collection procedure (or research protocol) was conducted following the Systematic Search Flow (SSF) method described by Ferenhof and Fernandes [21]. The first step consists of formulating the search strategy, followed by querying the selected databases. The selected databases comprise ACM Digital Library, Scopus, IEEE Xplore, Engineering Village (Compendex), and ScienceDirect. These sources were chosen to ensure comprehensive coverage of peer-reviewed journal articles in engineering, computer science, and applied artificial intelligence, which are core domains for research on Prognostics and Health Management. IEEE Xplore and ACM Digital Library provide strong coverage of AI- and data-driven methods, while Scopus and Engineering Village offer broad multidisciplinary indexing and citation tracking. ScienceDirect complements this set by covering leading journals in mechanical, manufacturing, and energy engineering. These steps are iterative, aiming to refine the search string that retrieves the largest number of relevant publications. Subsequently, the most relevant databases were selected, and the strings were adjusted as necessary, without changing the core search terms.

This phase also considered specific inclusion criteria. For the present study, the selected materials had to originate from scientific journals, be written in English, and not be literature review articles. Table 2 presents the selected databases, specific search strings, imposed restrictions, and the number of publications retrieved from each source. The data collection was performed on May 15, 2025, with no temporal restriction on the date of publication.

The choice of data processing architecture plays a critical role in the practical implementation of Prognostics and Health Management systems, directly affecting latency, scalability, energy consumption, and data security. In the context of Industry 4.0, cloud-, edge-, and fog-based architectures represent the main paradigms adopted to process and distribute data generated by connected industrial assets.

Table 1 summarizes the main characteristics of cloud, edge, and fog computing architectures and provides a reference framework for understanding the processing interfaces later analyzed through the Morphological Matrix and the Cross-Consistency Matrix.

The collected articles were then consolidated and processed. Initially, the dataset consisted of 692 articles. The first filter involved the removal of duplicate entries, which was conducted by analyzing the article titles and Digital

Object Identifiers (DOI). Next, publications categorized as reviews or surveys were excluded. Subsequently, titles and abstracts were examined to remove studies related to unrelated domains. For the third filter, articles older than two years and with less than 10 citations per year were discarded. The exclusion of conference papers and the adoption of citation-based thresholds were deliberate methodological choices aimed at ensuring the maturity and reliability of the selected studies. Journal articles typically present more complete methodological descriptions, extended validation, and consolidated results when compared to conference publications. In addition, the use of a minimum citation rate was adopted to prioritize influential and widely recognized contributions, while maintaining the focus on recent developments representative of the current state of the art. Similar criteria have been adopted in previous systematic reviews to balance relevance, impact, and timeliness of the analyzed literature.

Finally, the remaining articles were read in full and classified. Papers that did not align with the topic of interest were removed, resulting in a refined knowledge base that constitutes the state of the art on the subject. Following the described collection procedure and the applied filters, a total of 18 articles were selected to form the bibliographic analysis portfolio and support the state-of-the-art review on the integration of AI and PHM. These articles are listed in Table 3, sorted by citation rate (citations per year).

To visualize the relevance and temporal distribution of the selected studies, the references were imported into the bibliometric analysis software Litmaps™. As shown in Fig. 3, each article is represented by a node positioned according to its year of publication (horizontal axis) and

Table 3 Selected articles for the systematic literature review

ID	Author (year)	Title	DOI	Citations
01	Qu et al. (2019)	A neural-network-based method for RUL prediction and SOH monitoring of lithium–ion battery	https://doi.org/10.1109/ACCESS.2019.2925468	363
02	Booyse et al. (2020)	Deep digital twins for detection, diagnostics and prognostics	https://doi.org/10.1016/j.ymssp.2019.106612	284
03	Soualhi et al. (2014)	Prognosis of bearing failures using hidden Markov models and the adaptive neuro-fuzzy inference system	https://doi.org/10.1109/TIE.2013.2274415	234
04	Ma et al. (2017)	Discriminative deep belief networks with Ant Colony optimization for health status assessment of machine	https://doi.org/10.1109/TIM.2017.2735661	147
05	Ren et al. (2021)	Cloud-edge-based lightweight temporal CNNs for RUL prediction in IIoT	https://doi.org/10.1109/JIOT.2020.3008170	146
06	Hua et al. (2021)	RUL prediction of PEMFC systems under dynamic operating conditions	https://doi.org/10.1016/j.enconman.2021.113825	139
07	Che et al. (2019)	Combining multiple deep learning algorithms for PHM of aircraft	https://doi.org/10.1016/j.ast.2019.105423	136
08	Soualhi et al. (2019)	Health monitoring of bearing and gear faults by using a new health indicator extracted from current signals	https://doi.org/10.1016/j.measurement.2019.03.065	106
09	Li et al. (2019)	Fault diagnosis of rotating machinery using DBN and 1D CNN	https://doi.org/10.1109/ACCESS.2019.2953490	93
10	Zhang et al. (2020)	RUL estimation Using LSTM neural networks and deep fusion	https://doi.org/10.1109/ACCESS.2020.2966827	76
11	Lin (2021)	ML application using Gaussian SVM for motor bearing fault diagnosis	https://doi.org/10.3390/electronics10182266	70
12	Wang et al. (2023)	RUL prediction of bearings using attention-based CNN-TCN architecture	https://doi.org/10.1109/ACCESS.2023.3255891	35
13	Cabrera et al. (2023)	Adversarial fault detector guided by one-class learning for centrifugal pumps	https://doi.org/10.1109/TMECH.2022.3226347	11
14	Behera et al. (2023)	GAN-based multi-task learning approach for PHM in IIoT	https://doi.org/10.1109/TASE.2023.3267860	11
15	Wu et al. (2023)	RUL prediction via data-driven deep learning fusion model (CALAP)	https://doi.org/10.1109/ACCESS.2023.3322733	6
16	Allal et al. (2024)	Efficient health indicators for RUL prediction of PEMFCs	https://doi.org/10.1016/j.ecmx.2023.100503	6
17	Zheng et al. (2023)	Gearbox fault diagnosis in edge-IoT based on CNN and multiwavelet transform	https://doi.org/10.3390/s23218669	4
18	Xu et al. (2024)	Multi-resolution LSTM-based RUL model for aero-engines	https://doi.org/10.1109/TVT.2023.3319377	0

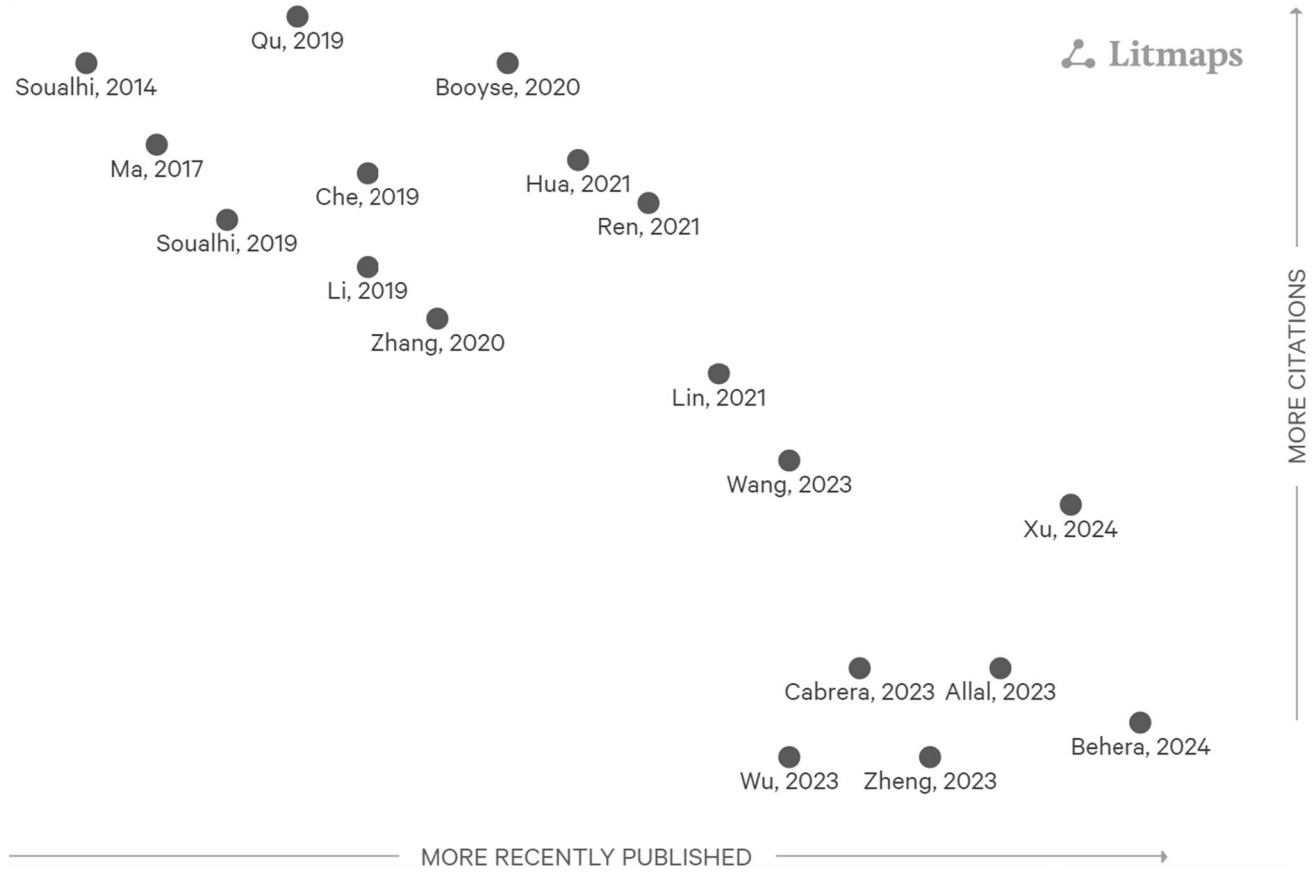


Fig. 3 Cross-consistency matrix (CCM)

number of citations (vertical axis). The resulting map highlights key contributions such as Soualhi et al. [65], Ma et al. [40], and Qu et al. [53] as earlier and highly cited works, indicating foundational studies in the field. More recent publications such as Behera et al. [6] and Xu et al. [81] appear on the right, reflecting current research directions but with lower citation counts due to their recency. This visualization provides a clear overview of the evolution of research interest and the scholarly impact of each article over time, reinforcing the relevance and diversity of the selected literature in the context of AI applied to Prognostics and Health Management.

4.2 Morphological matrix

The Morphological Matrix (MM) was applied due to its advantages in adaptability and suitability for comparing content from different areas of knowledge. According to Ritchey [57], this method is focused on structuring the full spectrum of relationships within complex, multidimensional, and non-quantifiable problems. This approach organizes the collected body of knowledge for analysis by defining dimensions and variants, resulting in a matrix capable of capturing the current state of the field and revealing unexplored future possibilities of a system [24].

The analysis is conducted after the selection of the most relevant articles. During the full-text analysis, the main themes addressed by the publications are identified and classified. Broad thematic categories are defined as dimensions, while specific topics are referred to as variants within each corresponding dimension. Throughout the review, four dimensions emerged as the most relevant:

- Dimension 1 – PHM aspect addressed.
- Dimension 2 – Applied AI/ML technique.
- Dimension 3 – Analyzed element.
- Dimension 4 – Data processing interface.

Table 4 presents the classification of each selected article according to the four dimensions. It also shows that most authors focused on Remaining Useful Life (RUL) estimation. Only two studies addressed both diagnostic and prognostic aspects, and approximately 50% of the research focused specifically on RUL estimation for bearings.

Table 4 Morphological matrix of selected studies

ID	Author (year)	PHM aspect (Dim. 1)	AI/ML technique (Dim. 2)	Analyzed element (Dim. 3)	Data processing interface (Dim. 4)
01	Qu et al. (2019)	RUL & HS	PA-LSTM	Lithium batteries	–
02	Booyse et al. (2020)	RUL	DDT GAN	Bearings	–
03	Soualhi et al. (2014)	RUL	ANFIS, HMM, AAC	Bearings	–
04	Ma et al. (2017)	HS	DDBN, ACO	Bearings	–
05	Ren et al. (2021)	RUL	LTCN	Bearings	–
06	Hua et al. (2021)	RUL	NN, ESN	Fuel cell	–
07	Che et al. (2019)	RUL	LSTM, DBN	Turbine	–
08	Soualhi et al. (2019)	HS	ANFIS	Bearings and gears	–
09	Li et al. (2019)	RUL	DBN, CNN, RBM	Bearings and gears	–
10	Zhang et al. (2024)	RUL	LSTM-FUSION	Turbine	–
11	Lin (2021)	RUL	SVM	Bearings	–
12	Wang et al. (2023)	RUL	CAM-TCN	Bearings	–
13	Cabrera et al. (2022)	HS	SVM, GAN-based	Centrifugal pump	–
14	Behera et al. (2024)	RUL & HS	LS-R-ACGAN	Turbine	IIoT
15	Wu et al. (2023)	RUL	CALAP	Turbine	IoT
16	Allal et al. (2024)	RUL	Random Forest	Fuel cell	–
17	Zheng et al. (2023)	HS	LMWT-MCNN	Bearings and gears	–
18	Xu et al. (2023)	RUL	MR-LSTM	Turbine	–

RUL Remaining useful life, *HS* Health state, *PHM* Prognostics and health management, *CNN* Convolutional neural network, *DBN* Deep belief network, *GAN* Generative adversarial network, *LSTM* Long short-term memory

4.2.1 Cross-consistency matrix (CCM)

The Cross-Consistency Matrix (CCM) is an $n \times n$ matrix constructed based on predefined dimensions and variants. In this study, $n = 15$. The horizontal axis of the matrix represents Dimensions 1, 2 and 3, while the vertical axis displays Dimensions 2, 3 and 4. This configuration yields 96 potential unique bi-dimensional variant pairings.

Each cell of the matrix corresponds to the interaction between two variants and is populated based on the classification of the selected articles. Each research paper was analyzed and mapped to the relevant cells according to its content. Therefore, each cell represents a specific interaction between variants, highlighting the explored areas and literature gaps [24].

It is important to note that the CCM does not account for any causal relationships or forms of association between the variants. As shown in Fig. 4, the resulting matrix reveals that only 46 out of the 96 theoretically possible interactions have been explored in the literature, which corresponds to 48% of the total. Consequently, 52% of the variant space remains unexplored. Additionally, the matrix highlights a high concentration of studies involving the use of CNN and LSTM techniques for predicting the Remaining Useful Life (RUL) of bearings. Only 2 out of the 96 papers included some form of research related to IoT or IIoT, and none of them addressed aspects related to AIoT.

The selection of the four morphological dimensions was guided by the core building blocks of Prognostics and Health Management systems and by the need to capture both methodological and implementation-oriented aspects of AI-enabled PHM solutions. Dimension 1 (PHM aspect) distinguishes whether studies address health state assessment, remaining useful life estimation, or both, reflecting the fundamental functional objectives of PHM. Dimension 2 (AI/ML technique) represents the analytical methods employed to extract information from data, encompassing both deep learning and classical machine learning approaches. Dimension 3 (analyzed element) characterizes the physical system or component under monitoring, which directly influences signal availability, degradation mechanisms, and data characteristics. Dimension 4 (data processing interface) captures where and how data are processed, ranging from centralized cloud-based architectures to distributed and edge-oriented solutions, which is critical for assessing practical deployability in industrial environments (Table 5).

5 Analysis of the cross-consistency matrix

Based on the concepts presented and the literature collected, classified, and analyzed, it is possible to delineate the state of the art concerning the integration of the studied themes and their practical applications. The following subsections detail the main application areas and advances achieved by researchers in the field.

5.1 Condition monitoring based on AI (Dim.1: Var.2 vs. Dim.2)

Among the papers collected, processed, and classified, only six were proposed, these include the works of Qu et al. [53], Ma et al. [40], Soualhi et al. [66], Cabrera et al. [11], Li et al. [37], Behera et al. [6], and Zheng et al. [83].

Fig. 4 Cross-consistency matrix (CCM)

		Dim.1		Dim.2								Dim.3				
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Dim.2	3	4	1													
	4	5	1													
	5	2	1													
	6	2	1													
	7	2	2													
	8	1	1													
	9	1	1													
	10	1														
Dim.3	11	6	2	3	1	2		1	2	1						
	12	1	2	1					1							
	13	3	1		1		1				1					
	14	3			4	2	1	1								
	15		1					1		1						
Dim.4	16		1	1									1			
	17						1	1							1	

Table 5 Variants identified by dimension

Dim. 1	Dimension 2	Dimension 3	Dimension 4
V1 – RUL	V3 – LSTM	V11 – Bearings	V16 – IoT
V2 – HS	V4 – CNN	V12 – Gears	V17 – IIoT
	V5 – DBN	V13 – Batteries and fuel cells	
	V6 – RNN	V14 – Turbines	
	V7 – GAN	V15 – Centrifugal pump	
	V8 – Fuzzy[1]		
	V9 – SVM[2]		
	V10 – Random Forest[3]		

[1]Fuzzy logic is a branch of logic that deals with uncertainty and imprecision. Unlike traditional Boolean logic, where variables are binary, fuzzy logic allows variables to take values between 0 and 1, representing degrees of membership in a set

[2]Support Vector Machine (SVM) is a supervised learning method commonly used for classification and regression tasks

[3]Random Forest is a widely used machine learning method for classification and regression, based on decision tree ensembles

Qu et al. [53] presented research involving lithium-ion batteries, leveraging the well-established correlation between battery capacitance and the degree of degradation, known as State of Health (SoH), to estimate the current condition of the component.

The studies conducted by Ma et al. [40], Zheng et al. [83], and Soualhi et al. [66] focused on the health estimation of rolling-element bearings. These studies utilized vibration data collected via accelerometers, with reported sampling frequencies of 10 kHz, 66.7 kHz, and not declared, respectively. Regarding centrifugal pump systems, Cabrera et al. [11] based their condition monitoring strategy solely on vibration signals, using these to assess the current health state of the system.

Li et al. [37] proposed an integrated approach combining one-dimensional Convolutional Neural Networks (1D-CNN) and Deep Belief Networks (DBN). In their work, vibration data from bearings were collected on a test bench and analyzed to estimate system condition (HS) and classify fault types. As illustrated in Fig. 5, the proposed method achieved an accuracy of 98.17% in the experiments performed by the authors.

Finally, Behera et al. [6] conducted their analyses using the simulation software developed by NASA, known as the Commercial Modular Aero-Propulsion System Simulation (C-MAPSS). As such, no real-world data collection related to health state (HS) was performed in their study. From this, it can be concluded that among the publications addressing HS, all analyses were based on a single type of measurement, either vibration or capacitance. Therefore, studies exploring correlations between multiple signals or multimodal sensing approaches were not identified.

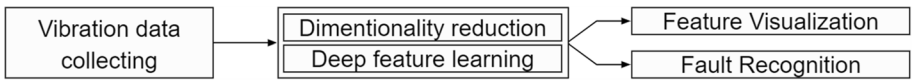


Fig. 5 Architecture proposed by Li et al. (2019) combining 1D-CNN and DBN for HS estimation. Source: Adapted from Li et al. (2019)

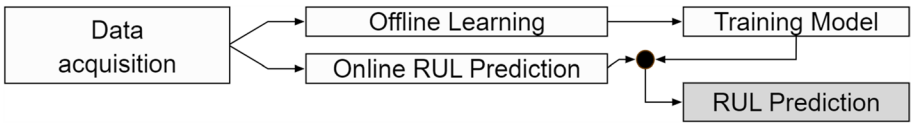


Fig. 6 Architecture of CAMTCN model proposed by Wang et al. (2023). Source: Adapted from Wang et al. (2023)

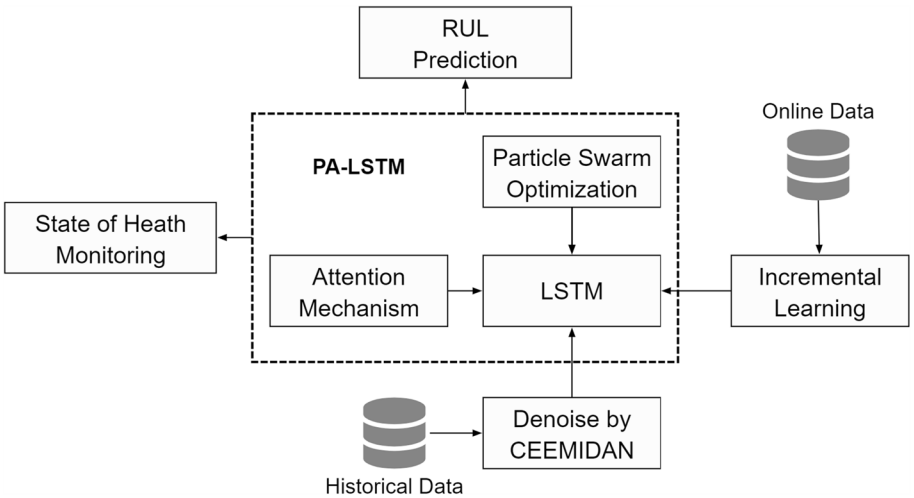


Fig. 7 Framework proposed by Qu et al. [53] combining LSTM and PSO for HS and RUL estimation

5.2 Remaining useful life (RUL) determination based on AI (Dim.1: Var.1 vs. Dim.2)

The prediction of Remaining Useful Life (RUL) was the quadrant with the highest number of related studies, with a total of 31 articles. Of these, five focused on the application of Convolutional Neural Networks (CNN), four addressed Long Short-Term Memory networks (LSTM), and the remaining were evenly distributed among other techniques.

During the review process, some studies employing traditional prediction techniques that were not based on AI advancements were excluded from the analysis. These included methods such as Fuzzy logic (V8), Support Vector Machines (SVM) (V9), and Random Forests (V10). CNN emerged as the most widely adopted technique, with relevant studies by Ren et al. [56], Wang et al. [77], Allal et al. [4], and Wu et al. [80].

The study by Ren et al. [56] mentioned the potential for IIoT integration, although no practical implementation was presented. The authors compared LSTM, GRU, and LTCN as predictive models. They introduced the distinction between single-time (ST) and multi-time (MT) sampling structures within the LTCN framework. Using bearing vibration datasets, their analysis showed that LTCN outperformed the other models in terms of Root Mean Square Error (RMSE) and required less data processing time compared to LSTM.

Bearing life prediction based on vibration signals was also addressed in the development reported by Wang et al. [77], who proposed a method called Convolutional Attention Mechanism and Temporal Convolutional Network (CAMTCN). In this approach, multichannel signals are input, normalized, and processed by a prediction algorithm, followed by the application of an optimizer and backpropagation to minimize error and define the optimal network structure. Figure 6 illustrates the architecture proposed by the authors. According to the study, the method demonstrated full-cycle vibration data processing capabilities, although it did not consistently outperform non-integrated methods in RMSE performance.

The use of the Random Forest (RF) method was addressed in the study by Allal et al. [4], in which fuel cell data were collected from standardized datasets. The results presented in the publication emphasize that less complex methods such as RF are still capable of effectively analyzing data without the high computational demand typically required by neural network models. According to the authors, this performance is attributed to the nature of the data being analyzed, highlighting that a deep understanding of data characteristics is fundamental for achieving accurate analysis.

LSTM was identified as the second most frequent technique among the reviewed articles, with studies published by Qu et al. [53], Che et al. [13], Zhang et al. [82], and Xu et al. [81]. All of these studies utilized standardized datasets and did not report real-world experimental implementation of their models.

In the work of Qu et al. [53], both health state (HS) and RUL estimation were addressed for lithium-ion batteries. The parameter used was the State of Health (SoH), evaluated via battery capacitance. The proposed framework, illustrated in Fig. 7, involves real-time measured values used for incremental learning, while historical data are preprocessed through noise filtering before being fed into the LSTM model. The Particle Swarm Optimization (PSO) technique was applied to optimize the weights in the model.

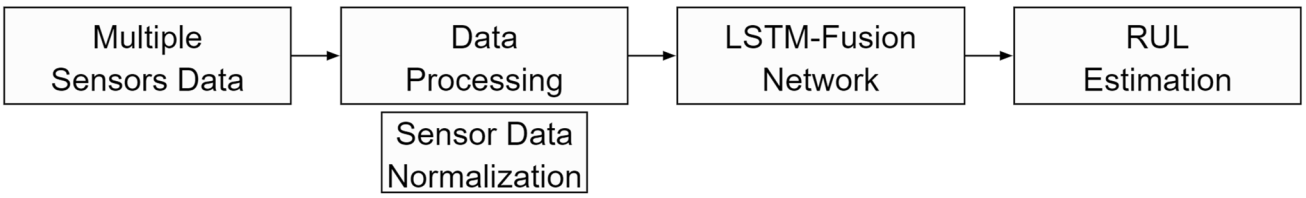


Fig. 8 LSTM-fusion architecture proposed by Zhang et al. (2020)

The proposed model by Qu et al. [53], referred to as PA-LSTM, integrates Particle Swarm Optimization (PSO), Long Short-Term Memory (LSTM), and a noise filtering stage. According to the authors, this combined approach achieved lower Root Mean Square Error (RMSE) values compared to standalone LSTM and basic RNN models. The publications by Che et al. [13], Zhang et al. [82], and Xu et al. [81] all used the same standardized dataset for turbomachinery, known as C-MAPSS. In Che et al. [13], the authors proposed a parallel application combining Deep Belief Networks (DBN) and LSTM. In their framework, the DBN is responsible for estimating the health state (HS), while the LSTM model handles RUL prediction based on sensor data inputs.

Zhang et al. [82] introduced a method called LSTM-Fusion for RUL estimation. In this approach, input data obtained at different sampling rates or time windows are passed through n parallel LSTM layers before undergoing high-level concatenation, as illustrated in Fig. 8. This architecture demonstrated improved predictive accuracy over standard LSTM models and enabled the incorporation of asynchronous data. However, the approach also required greater computational resources compared to the other methods evaluated.

Finally, Xu et al. [81] proposed a method similar to the one developed by Zhang et al. [82], but with the addition of Discrete Wavelet Transform (DWT) for noise removal. However, the results obtained were inferior to those of the LSTM-Fusion model.

5.3 Critical comparison of AI/ML techniques for PHM applications

Beyond the frequency of use identified in the Morphological and Cross-Consistency Matrices, the reviewed studies reveal important trade-offs among different AI and ML techniques when applied to Prognostics and Health Management. Deep learning approaches, particularly Convolutional Neural Networks and Long Short-Term Memory networks, dominate the literature due to their ability to automatically extract complex features and model temporal degradation patterns. However, these methods typically require large volumes of labeled data, substantial computational resources, and extensive training efforts, which may limit their applicability in industrial environments with constrained data availability or real-time processing requirements.

In contrast, classical machine learning techniques such as Support Vector Machines and Random Forests appear less frequently in recent publications but offer advantages in terms of interpretability, lower computational cost, and robustness when operating with smaller datasets. These characteristics make such methods attractive for early-stage deployments, legacy systems, or applications where transparency and explainability are critical for maintenance decision-making. Hybrid approaches, combining deep learning models with physics-based indicators or traditional classifiers, have also emerged as promising alternatives to balance predictive performance and practical feasibility.

From an implementation perspective, the choice of AI/ML technique is strongly influenced by deployment constraints. Deep architectures are often better suited to centralized or cloud-based processing, whereas lightweight models and feature-based methods are more compatible with edge or embedded platforms. Despite this, the Cross-Consistency Matrix indicates that few studies explicitly address these trade-offs or report real-world validation under industrial constraints, highlighting a gap between algorithmic performance reported in benchmark datasets and the requirements of scalable, field-deployable PHM solutions.

5.4 Practical implementation, validation, and industrial maturity of PHM solutions

Although a wide range of AI-based approaches has been proposed for Prognostics and Health Management, the results of the Morphological and Cross-Consistency Matrices indicate that most studies remain at a conceptual or laboratory validation level. The predominance of benchmark datasets and simulated environments suggests limited progression toward higher technology readiness levels, where models are validated under real operational conditions and integrated into industrial workflows.

Validation strategies reported in the literature are largely focused on predictive accuracy metrics, such as RMSE or MAE, often evaluated offline and under controlled conditions. While these metrics are essential for algorithmic comparison, they provide limited insight into robustness,

reliability, and long-term performance in real-world applications. Aspects such as sensor degradation, data drift, changing operating regimes, and maintenance-induced disturbances are rarely addressed, despite their relevance for industrial deployment.

From an implementation perspective, practical constraints related to data availability, communication bandwidth, cybersecurity, and computational resources play a decisive role in determining feasible PHM architectures. The limited number of studies involving IoT, IIoT, or edge-based processing identified in this review highlights the gap between high-performance predictive models and their integration into scalable, field-deployable systems. Addressing these challenges requires validation frameworks that go beyond algorithmic accuracy and explicitly consider system-level performance, deployment constraints, and operational reliability.

5.5 Cross-analysis of PHM aspect, target system, and interface (Dim.1 × Dim.3 × Dim.4)

This cross-analysis between dimensions results in a total of 24 possible combinations, of which only 10 were found to be addressed in the literature. Studies involving IoT and IIoT applications were presented exclusively by Zheng et al. [83], Behera et al. [6], and Ren et al. [56].

Most of the reviewed publications relied on standardized datasets. Notably, only Cabrera et al. [11] addressed PHM for centrifugal pumps based on vibration data. No studies were identified that explored the relationship between processing capacity, data storage, and data transmission in the PHM context.

5.6 Interface for PHM application (Dim.1 vs. Dim.4)

As identified in the table and figure, only two out of the eighteen reviewed publications presented physical interfaces. Zheng et al. [83] employed IoT devices to collect data at the component level - e.g., gears and bearings - followed by the transmission of readings to the edge level for processing. This communication was unidirectional, with no return updates or modifications at the device level. Behera et al. [6] mentioned the benefits of IIoT technology in conjunction with PHM, however, no practical implementation was reported, as the study relied on simulation data. The remaining studies were based on standard databases or computer-based environments.

When interpreted in light of the processing architectures summarized in Table 1, the results of the Cross-Consistency Matrix indicate a clear underrepresentation of edge- and fog-based PHM solutions. This suggests that, despite their advantages in terms of latency and real-time decision-making, distributed processing architectures remain largely unexplored in current AI-enabled PHM research.

5.7 Link between MM/CCM findings and identified research gaps

The research gaps highlighted in this study arise directly from the structured evidence obtained through the Morphological Matrix (MM) and the Cross-Consistency Matrix (CCM). The MM revealed a strong concentration of studies within a limited subset of variant combinations, particularly the use of CNN and LSTM methods applied to RUL prediction of bearings, while several other combinations remain scarcely explored or entirely absent.

This observation is reinforced quantitatively by the CCM results, which show that only 46 out of the 96 theoretically possible variant interactions (48%) are present in current literature. The remaining 50% represent unexplored regions of the design space where no study has combined PHM aspects, AI/ML techniques, analyzed elements, or processing interfaces.

A notable finding is the near absence of research involving IoT, IIoT, or AIoT interfaces (Dimension 4). Only two studies partially address these variants, and none explore bi-directional communication, embedded intelligence, or real-time edge processing. Similarly, the MM indicates that almost all HS-related studies rely on a single sensing modality, revealing a lack of multimodal or multi-sensor approaches.

These empirical patterns form the foundation of the research gaps presented in Sect. 5.5. Each identified gap corresponds directly to an unexplored or weakly represented interaction revealed by the MM/CCM analysis. This explicit alignment ensures that the proposed research agenda is grounded in objective, systematically derived evidence rather than subjective interpretation.

5.8 Identified research gaps

Based on the analyses conducted and the topics discussed in the theoretical framework, several research gaps and opportunities have been identified. The growing relevance of this subject is driven by the increasing complexity of industrial systems, the demand for predictive and data-driven maintenance strategies, and the broader integration of digital technologies within the context of Industry 4.0. These factors have contributed to the expansion of interest in Prognostics and Health Management (PHM), particularly in applications involving intelligent monitoring and decision-making support. The following research gaps and opportunities have been identified:

- Application and prediction methods for HS and RUL in systems, equipment, or components under real-world conditions, beyond standard benchmark datasets;
- AIoT applications in which processing is performed *in loco*, aiming to increase data reliability and security;
- AIoT applications in critical operations for condition monitoring (HS) and fault prediction to improve maintenance planning;
- Application in more complex systems, such as road vehicles, agricultural machinery, locomotives, or ships operating in regions with limited data bandwidth;
- Methodology for the development, verification, or validation of the proposed method and equipment;
- Use of data already collected through sensing and available in standardized data buses to perform fault predictions;
- Identification of criteria and systematization of procedures for selecting equipment for AIoT deployment in real field applications;
- Definition of criteria for data integration into intelligent models, such as in the context of Smart Farming.

5.9 Artifacts identified in the state of the art

Throughout the analyses, artifacts or constructs were identified and recorded, and the content was organized as follows:

- Definition of the subsystem or component of interest
  - Soualhi et al. [66] highlights the need for architectural and functional analysis to identify the critical component.
- Data collection and preprocessing
  - Ren et al. [56] applied data preprocessing at the edge level, combined with cloud processing;
  - Cabrera et al. [11] employed the *Signal Augmentation* technique[1] to address issues arising from limited data availability;
- Evaluation of results
  - Qu et al. [53], Booyse et al. [9], Ren et al. [56], Behera et al. [6], Xu et al. [81], and Wu et al. [80] based their result evaluation on the RMSE metric,[2];
  - Hua et al. [27] and Allal et al. [4], in addition to using RMSE to assess accuracy, also reported the coefficient of determination (R-squared);
  - Wang et al. [77] presented the analysis based on both MAE,[3] and RMSE;
- Identification of signals and parameters
  - Soualhi et al. [66] proposed a function decomposition flow, from the system level down to data acquisition, referencing fault tree analysis and failure mode and effects analysis, though no further detail is provided regarding the process.
- Dissemination of results
  - Qu et al. [53] proposes a framework in which SOH,[4] is quantified, and an alert is triggered when the calculated value falls below a predefined threshold, though the criterion for setting this threshold is not mentioned;
- Application of advanced data analysis and prediction techniques for HS and RUL in real-world systems, equipment, or components, beyond standardized datasets;
- Integration of PHM in complex systems, such as road vehicles, agricultural machinery, locomotives, or maritime vessels operating in areas with limited data bandwidth;
- Development of methodologies for validation, verification, or testing of PHM models and embedded hardware;
- Use of sensor data already available on standardized communication buses for failure prediction;
- Identification of criteria and systematization of procedures for selecting equipment suitable for AIoT integration in real-field applications.

5.10 Limitations

This review presents some inherent limitations. First, only peer-reviewed journal articles were included, excluding conference papers, technical reports, and grey literature that may contain relevant insights. Second, the Morphological Matrix and Cross-Consistency Matrix depend on predefined dimensions and variants; alternative classifications could yield different interaction patterns. Moreover, most selected studies rely on benchmark datasets rather than real industrial data, which may bias the identified trends. Finally, due

[1] *Signal Augmentation* is a commonly used technique for generating new data based on real samples, mainly applied during the training phase of models.

[2] RMSE, or *Root Mean Squared Error* is a measure of absolute error in a dataset.

[3] MAE, or *Mean Absolute Error* measures the average absolute error between predicted and actual values.

[4] SOH, or *State of Health* is a health condition quantification metric, similar to HS, commonly used to assess battery storage capacity.

to the heterogeneity of datasets, AI models, and evaluation metrics, a quantitative meta-analysis was not feasible, and conclusions should be interpreted within the context of qualitative evidence synthesis.

The adoption of temporal and citation-based filtering criteria may introduce selection bias by excluding recent publications that have not yet accumulated citations or older studies with lower citation rates despite their conceptual relevance. However, this trade-off was considered acceptable given the objective of capturing a representative snapshot of mature and impactful research at the intersection of artificial intelligence and Prognostics and Health Management. Future studies could extend the analysis by incorporating conference papers, grey literature, or alternative impact indicators to further broaden the scope of the review.

6 Conclusion

This systematic review investigated the application of artificial intelligence and machine learning techniques to Prognostics and Health Management in critical machinery through an integrated SSF–MM–CCM framework. By structurally decomposing the literature into dimensions related to PHM aspects, analytical techniques, target systems, and processing interfaces, the study enabled a quantitative assessment of current research trends and gaps that extends beyond conventional descriptive reviews.

The results indicate a strong concentration of studies on deep learning approaches, particularly CNN- and LSTM-based models applied to remaining useful life prediction of bearings and other isolated components. In contrast, complex systems, multimodal sensing strategies, distributed processing architectures, and real-world industrial deployments remain significantly underrepresented. The Cross-Consistency Matrix analysis revealed that less than half of the theoretically possible combinations among the defined dimensions have been explored, highlighting substantial opportunities for future research.

From a practical perspective, the findings suggest a persistent gap between high predictive performance reported in laboratory settings and the requirements of scalable, robust, and deployable PHM solutions. Limitations related to data availability, validation under operational conditions, and integration with industrial communication and processing infrastructures continue to constrain the transition from experimental models to field-ready applications.

By explicitly mapping these gaps, this review contributes a structured foundation for guiding future investigations toward underexplored combinations, particularly those involving edge intelligence, AIoT architectures, system-level validation, and long-term operational assessment. These directions are essential for advancing AI-enabled PHM from algorithm-centric developments toward reliable and industrially viable maintenance solutions.

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Data availability Data will be made available on reasonable request.

Declarations

Conflict of interest A preliminary and abridged version of this work was presented and published at the Proceedings of the 27th International Congress of Mechanical Engineering (COBEM 2025). The present manuscript is a substantially extended and fully revised version tailored to the standards of journal publication. The authors declare that there are no known financial interests or personal relationships that could have inappropriately influenced the conduct or findings of this research. All authors have reviewed and approved the manuscript.

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